

BACHELOR'S THESIS

Exploring Hyperbolic Embeddings of Finite Metric Spaces

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Abstract

Abstract der Thesis

Zusammenfassung

Hier die Zusammenfassung

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1 Introduction

1.1 Overview and Motivation

This section gives an overview about prerequisites from math and computer science (complex numbers, vector spaces, gradients, data, automatic differentiation, representation learning), followed by a motivation about embeddings, why they are useful in practice. WENIG FORMELN, FOKUS AUF MOTIVATION UND INTUITIVES VERSTÄNDNIS.

1.2 Organization

This section describes how the following thesis is organized and which problems are tackled in which chapter.

2 Background

2.1 Riemannian Geometry

In order to gain an understanding of the embedding space, the basics of Riemannian geometry are now explained.

2.1.1 Tangent Space

In the following a basic understanding of smooth manifolds is assumed. This section follows the introduction of tangent spaces $T_p M$, $p \in M$ in chapter 1 of [Car92]. At every point p of an arbitrary smooth manifold M , tangent vectors can be defined. For this, $\alpha : (-\epsilon, \epsilon) \rightarrow M$ denotes a smooth curve which holds $\alpha(0) = p \in M$, i.e. it evaluates to p at $t = 0$. Additionally $\mathcal{D}$ refers to the set of functions, which are differentiable at point p . A tangent vector w at p is the tangent vector to $\dot{\alpha}(0)$ defined as

$$\dot{\alpha}(0)f = \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0}, \quad f \in \mathcal{D}. \quad (2.1)$$

The set of all tangent vectors spans the tangent space $T_p M$ at point p . Geometrically this space can be interpreted as the linear approximation of M at p .

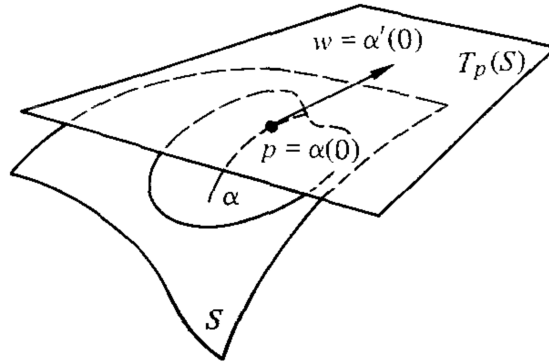


Figure 2.1: The tangent space of a surface S at point p [Car16]

2.1.2 Lengths and Metrics

While optimizing the embeddings, distances are computed in both the euclidean and the hyperbolic space. In the euclidean case, this is trivial using the euclidean metric, due to the flat space. In the hyperbolic case on the other hand, it's necessary to use some Riemannian metric. A Riemannian metric associates a smoothly varying symmetric and positive-definite bilinear form (inner product)

$$g(\cdot, \cdot) = \langle \cdot, \cdot \rangle_p : T_p M \times T_p M \rightarrow \mathbb{R} \quad (2.2)$$

to each point p on M [AMS09]. In practice, ds is also often used to denote the Riemannian metric. The Poincaré Disk model, which is used for the embedding space, can be considered a Riemannian manifold, which is a smooth manifold with an associated Riemannian metric [Car92]. Using the Riemannian metric of the manifold the length

$$L(\gamma) = \int_{t_0}^{t_1} \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt \quad (2.3)$$

of a curve $\gamma : [t_0, t_1] \rightarrow M$ can be computed [AMS09]. The Riemannian distance uses this, to denote the distance between two points $x, y \in M$

$$\text{dist}(x, y) = \inf_{\Gamma} L(\gamma) \quad (2.4)$$

[AMS09] as the length of the shortest curve γ from the set Γ of all curves which pass x and y . Those length-minimizing shortest paths $\gamma(t)$ are called geodesics.

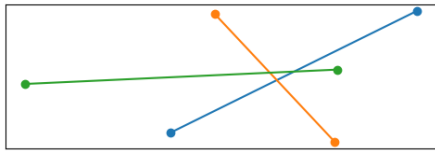


Figure 2.2: Geodesics in the euclidean plane

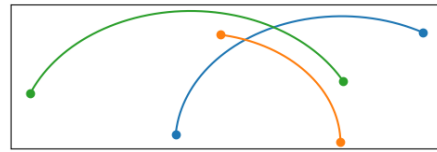


Figure 2.3: Geodesics in the hyperbolic Lobachevsky plane

2.1.3 The Exponential Map

During the optimization of the embeddings, data points in the embedding space are assigned to new points on the Riemannian manifold M . The direction and length of the

change depends on the Riemannian gradient $\text{grad } f$, which lies in the tangential space $T_p M$ of M at point p . The newly calculated value is assigned by the exponential map

$$\exp_p(v) := \gamma_{p,v}(1). \quad (2.5)$$

Geometrically speaking, it assigns a point $p \in M$ together with a tangent vector $v \in T_p M$ to a new point in M by evaluating the geodesic $\gamma(t)$ at $t = 1$ with initial velocity v at point p .

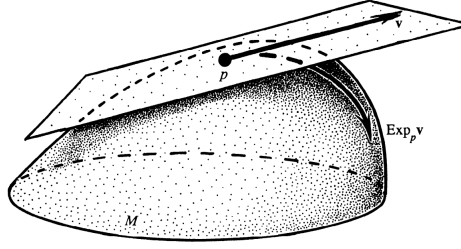


Figure 2.4: The exponential map at point p [Mor93]

2.2 Numerical Optimization on Manifolds

The embedding of the data follows an optimization approach. This means, instead of directly calculating good representations of the data in the embedding space, they are first randomly embedded and then continuously improved by minimizing a loss function $\mathcal{L}$. This chapter focuses on numerical optimization on manifolds in general, technical details will later be explained in chapter 3.2.

2.2.1 First Order Optimization

Before moving on to optimization on manifolds, consider the general optimization problem in the euclidean space

$$\inf_{x \in \mathbb{R}^n} f(x), \quad f \in \mathcal{F} \quad (2.6)$$

i.e. the objective function f from the set of functions $\mathcal{F}$ is getting minimized by finding optimal values of x . This can be achieved by moving x in the direction of the steepest descent $-\text{grad } f(x(t))$ until a critical point with $\text{grad } f(x(t)) = 0$ is reached. A numerical implementation is the gradient flow, which is the solution map $x_0 \rightarrow x(t)$ to the differential equation

$$\dot{x}(t) = -\text{grad } f(x(t)), \quad x(0) = x_0. \quad (2.7)$$

However, since this approach is impractical in most cases when using numerical methods [AMS09], the focus in the following is on algorithms

$$x_{t+1} = x_t + h_t \Delta x_t \quad (2.8)$$

which update x step wise in the search direction $\Delta x_t \in \mathbb{R}^n$ with step size $h_t \in \mathbb{R}$. In practice, this can be implemented algorithmically in the following way:

Algorithm 1 Basic optimization algorithm

Require: x_0

while stopping criterion not achieved **do**

1. Determine the search direction Δx_t

2. Find a suitable step size h_t

▷ Line search

3. Update $x_{t+1} = x_t + h_t \Delta x_t$

end while

More detailed information on the line search step can be found in chapter 2.2.3.

2.2.2 Riemannian Stochastic Gradient Descent

The following section explains the Riemannian stochastic gradient descent algorithm as described in [NK17] and [Bon13], which present an iterative algorithm for optimization on Riemannian manifolds. From now on $\theta_t \in M$ will be used to denote points on the manifold, η_t the learning rate at step t and $\mathcal{L}$ a loss function, which is the objective function that is getting minimized.

Algorithm 2 Riemannian stochastic gradient descent (RSGD)

Require: $\theta_0 \in M$

while stopping criterion not satisfied **do**

1. Compute $\text{grad } \mathcal{L}(\theta_t)$

2. Find a suitable step size η_t

▷ Line search

3. Update $\theta_{t+1} = \exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))$

end while

In the algorithm, the Riemannian gradient of $\mathcal{L}$ is first computed. It is the tangent vector in $T_{\theta_t}M$ that describes the direction of the steepest ascent. It is then rescaled in

length with a suitable step size η_t and inverted, such that it points in the direction of the steepest descent. To obtain the new θ_{t+1} the exponential map is used. Information on how this procedure is used in practice for the Poincaré disk is discussed in chapter 3.

2.2.3 Line Search - Accelerating RSGD

For fast convergence in RSGD, it's necessary to choose a suitable learning rate η_t in each step t . To ensure this in every iteration, an adaptive learning rate is used during the optimization, which leads in practice to exponential convergence in the neighborhood of local minima [Formelsammlung]. The simplest assumption how we can choose η_t is

$$\mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) < \mathcal{L}(\theta_t) \quad (2.9)$$

such that the loss decreases in each iteration. However, this condition is not enough to obtain convergence [NW06]. Instead, a sufficient decrease condition is needed to guarantee convergence. A condition that ensures this is the Armijo condition

$$\mathcal{L}(\theta_t) - \mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) \geq c \|\text{grad } \mathcal{L}(\theta_t)\|_g^2 \quad (2.10)$$

with $c > 0, \forall k \in \mathbb{N}$. Based on this condition, η is determined below in algorithm 3 using backtracking line search. This is an inexact line search algorithm that iteratively changes the step size until the Armijo condition is fulfilled. Inexact means that the optimal step size is not calculated, but an approximation that leads to a sufficient decrease, since calculating it exactly is too time-consuming in most cases.

Algorithm 3 Backtracking Line-Search

Require: $\theta_t \in M, \eta_t > 0, \tau, r \in (0, 1)$ (typical values : $\tau = \frac{1}{2}, r = 10^{-4}$)
while $\mathcal{L}(\theta_t) - \mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) < r\eta_t \|\text{grad } \mathcal{L}(\theta_t)\|_g^2$ **do**
 Update step size $\eta_t = \tau\eta_t$
end while
return η_t

3 Poincaré Disk and Metric Data Embeddings

3.1 Hyperbolic Geometry

Hyperbolic geometry is a non-Euclidean geometry, which means that not all Euclidean axioms are fulfilled. In fact the fifth Euclidean axiom, which can be easier to understand expressed as the parallel postulate

”Given a line and a point not on it, there is exactly one line going through the given point that is parallel to the given line” [Can+97]

is not satisfied. Instead there can exist multiple lines through the given point which are parallel to the given line.

3.1.1 The Poincaré Disk

A model of hyperbolic geometry that will be used from now on is the Poincaré disk. It’s the unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\} \quad (3.1)$$

with the associated Riemannian metric

$$ds = \frac{2|dz|}{1 - |z|^2}. \quad (3.2)$$

This forms a model of hyperbolic space with constant negative curvature -1. The boundary of the disk, which is the sphere S^1 , is denoted as $\partial\mathbb{D}$. It represents points that are infinitely distant away from the center [NK17].

Geometric properties

The Riemann metric ensures that Riemannian distances grow exponentially with increasing distance from the center. This leads to the fact that geodesics are only straight lines if they are diameters, otherwise they are an arc of the circle that is orthogonal to $\partial\mathbb{D}$.

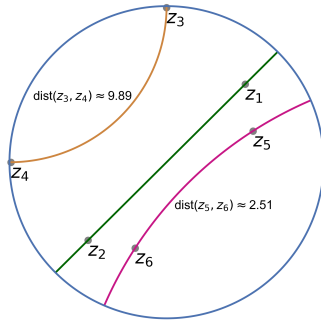


Figure 3.1: Geodesics in the Poincaré Disk

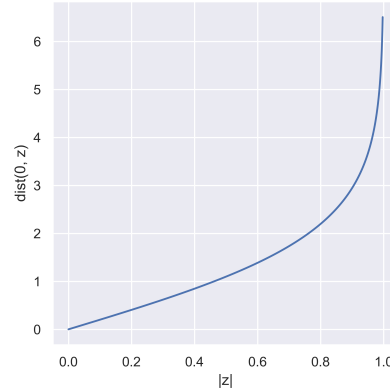


Figure 3.2: Growth of Riemannian distance in Poincaré Disk

Note in particular how the Riemannian distances in figure 3.1 differ by almost a factor of 4, although they would be very similar in Euclidean space.

3a - Shortest distances using Möbius transformations

First Möbius transformations in general as isometries. Then derive distances by first using Möbius transformations which maps one point to the center and then compute the distance from the center to the other point.

3b - Shortest distances using cross ratio

Use the cross ratio to compute distances and explain the idea geometrically.

4 - Exponential Map of the Poincaré Disk

Derive a formula for the exponential map, which is later used in RSGD.

3.2 Embedding Metric Data into the Poincaré Disk

4 Comparing Different Models of Hyperbolic Space - OPTIONAL

4.1 Existing Models of Hyperbolic Space

Poincaré half plane, Poincaré disk, Hyperboloid model, etc.

4.2 Evaluating the Models

5 Experiments and Discussion

5.1 Implementation Details

Implementation in torch or jax with focus on scalability and hardware acceleration

5.2 Evaluation on Toy and Real World Data

Results on toy data and real world data, correctness, distortion, runtime efficiency, memory efficiency

5.3 Conclusion

What went good, what wrong, what have we learned.

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